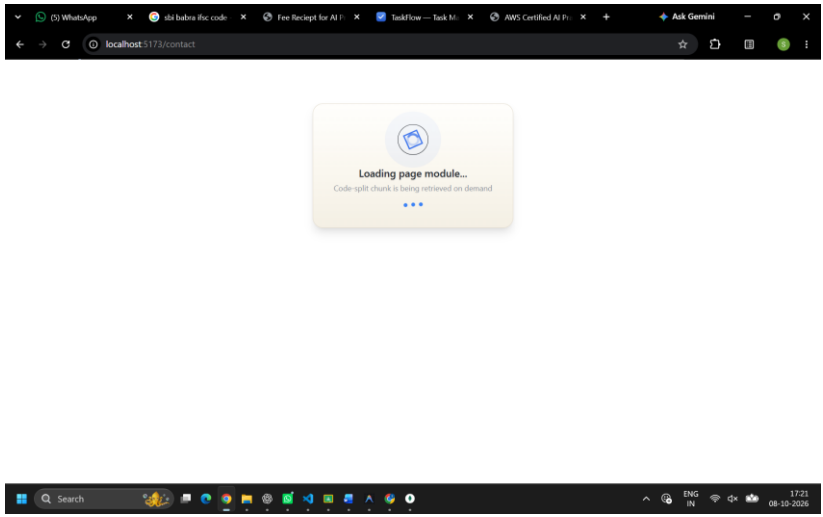


Practical 8:- Performance Optimization and Lazy Loading in React

Objective:	To improve frontend performance using lazy loading and code splitting techniques.
Problem Statement:	• Optimize the React frontend of the Task Management application by implementing lazy loading for route-based components. • Apply React.lazy() and Suspense to at least the Projects and Contact routes (or equivalent route-based components). • Provide a meaningful fallback UI while a lazy-loaded chunk is loading. • Measure and compare bundle size and load time before and after optimization using browser developer tools. • Document the before/after comparison with screenshots or recorded values.
Implementation	LazyLoading Spinner : <pre>import { Database } from 'lucide-react' const Spinner = ({ label = 'Loading repositories...' }) => (<div role="status" aria-live="polite" className="surface-raised flex flex-col items-center justify-center gap-5 px-8 py-16" > <Database className="size-5 animate-pulse text-bronze" /> <p className="text-sm font-medium text-ink-soft">{label}</p> Loading </div>) export default Spinner</pre> 

Outcome	Students will be able to apply code splitting and lazy loading in React to reduce initial bundle size, and will be able to measure and explain the resulting performance impact.
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